ANINDA ROY DHRUBA

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SUMMARY

CSE graduate from CUET and Lecturer at Northern University Bangladesh with research interests spanning Machine Learning, Reinforcement Learning, and optimization-driven Medical AI systems. Published work in hierarchical classification with custom loss functions, bias propagation in NLP, and time-series forecasting. Experienced with deep learning frameworks (PyTorch, TensorFlow) and motivated by interdisciplinary problems at the intersection of ML, optimization, and intelligent control specailly on Medical domain.

EDUCATION

Chittagong University of Engineering and Technology (CUET) **Aug 2025**
B.Sc. in Computer Science and Engineering CGPA: 3.89/4.00 · Dean's List

EXPERIENCE

Lecturer, Dept. of CSE **Aug 2025 – Present**
Northern University Bangladesh, Dhaka

- Teaching Data Structures, Structured Programming, OOP with Java, Computer Networks, Microprocessor & Assembly Language, and System Analysis & Design
- Designing course materials, assignments, and examinations

Research Associate **May 2025 – Present**
Learnify Research Lab

- Investigating 18 distinct bias types in NLP systems and their propagation mechanisms across model training pipelines and decoding strategies
- Survey paper published in Elsevier's Array Journal — Link

Graduate Researcher (Thesis) **Jun 2024 – Jul 2025**
CUET

- Designed hierarchical disease classification over 16 diseases and 4 specializations; formulated custom multi-objective loss functions to jointly optimize inter-level consistency and classification accuracy
- Fine-tuned domain-specific transformer models (BanglaBERT, ClinicalBERT); analyzed trade-offs between model complexity and predictive performance
- Portion of this work accepted at ICCIT Conference — Link

Research Collaborator **Mar – Jun 2025**
CUET

- Developed time-series forecasting models (LSTM, ARIMA, hybrid architectures) on 10-year PGCB electrical load data; achieved 1.09% MAPE via hyperparameter optimization and ensemble tuning
- Applied signal decomposition and feature engineering techniques to capture seasonal demand patterns

Industrial Attachment **Dec 2024 – Jan 2025**
RedDot Digital Limited, Dhaka

- Observed SDLC practices, project management workflows, and deployment strategies in industry

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, MATLAB
ML/DL & Optimization: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, BERT, GPT, CNNs, RNNs, LSTMs, Reinforcement Learning (Q-Learning, policy gradient), loss function design, hyperparameter optimization
Mathematical Foundations: Linear Algebra, Probability & Statistics, Convex Optimization, Time-Series Analysis
Web/Mobile: Next.js, Node.js, Flask, Flutter, HTML5/CSS3, RESTful APIs
Databases & Tools: MySQL, MongoDB, Firebase, Git, Docker, Linux, MATLAB (signal processing & simulation)

SELECTED PROJECTS

Hierarchical Medical Dialogue Classification — Formulated a multi-level classification pipeline over patient conversation data spanning 16 diseases; designed custom loss functions to enforce hierarchical consistency across specialization and disease layers. (PyTorch, BanglaBert) [*Thesis*]

Electrical Load Forecasting with Hybrid Time-Series Models — Built forecasting pipeline on 9-year grid data from PGCB; combined LSTM-based sequence models with ARIMA baselines and ensemble optimization to achieve 1.09% MAPE. Applied FFT-based decomposition for periodicity extraction. (Python, PyTorch, statsmodels)

Diabetic Retinopathy Detection — Developed a deep learning ensemble of 5 CNN architectures (VGG, ResNet, InceptionV3, DenseNet, EfficientNet) for multi-class retinal grading; analyzed inter-model agreement and uncertainty to guide ensemble weighting. Deployed via Flask interface. (TensorFlow, OpenCV) [GitHub]

Competitive Programming Duel Bot (Reinforcement-Style Ranking) — Designed a skill-matching and challenge system using Codeforces API; implemented an Elo-based rating update mechanism (analogous to TD-learning reward feedback) for adaptive opponent pairing and leaderboard management. (Python, Discord.py) [GitHub]

FreeToWork — Service marketplace with bidding system, user auth, and real-time notifications. (Next.js, MongoDB) [GitHub]

ACHIEVEMENTS & PUBLICATIONS

- **Elsevier Array Journal(Q1)** (2025) — Survey on bias types and propagation mechanisms in NLP systems [Link]
- **IEEE ICCIT 2025** — Hierarchical disease classification from patient dialogue using transformer models [Link]
- **Dean's List** — Maintained CGPA ≥ 3.75 throughout undergraduate studies at CUET
- **UNLP Shared Task 2024** — 5th place in two competition tracks (NLP shared task)
- **Competitive Programming** — 750+ problems solved on Codeforces, LeetCode, HackerRank
- **Creative Talent Search Competition (Math & ICT) 2019** — 1st place, Regional Round (Cumilla)